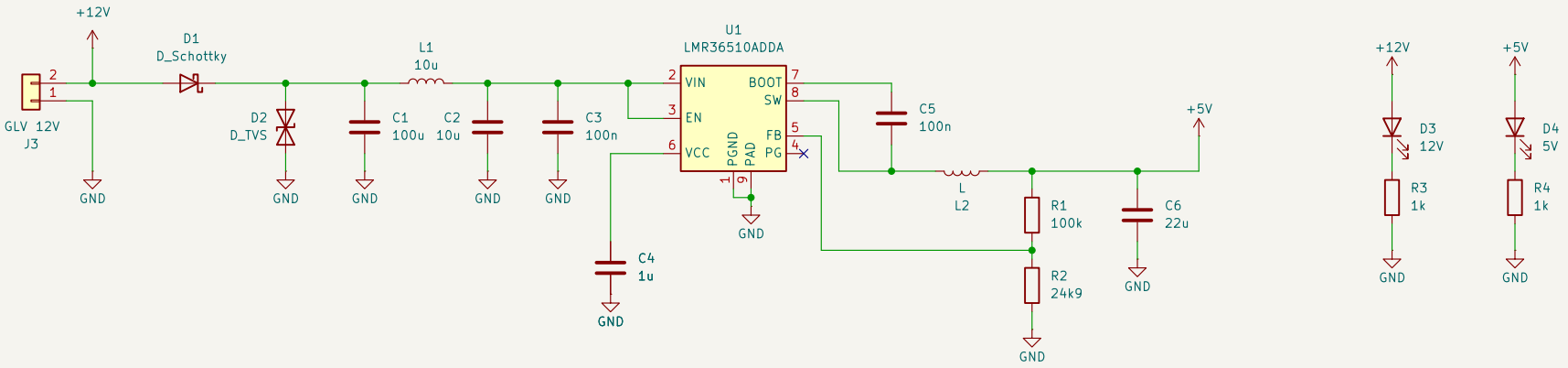


Repositório Monorepo: Hardware + Firmware + Docs
Barreira Galvânica Iso-SPI >= 2500V RMS (HMU2103NL)
Conformidade com Regulamento FSAE EV4.8 / EV8
ECU Central de Gerenciamento do Acumulador 20s18p
Formula ITA – Formula SAE Electric

Sheet: /
File: bms_carrier_shield.kicad_sch

Title: BMS Master Carrier Shield – NUCLEO-G474RE

Size: A3	Date: 2026-10-22	**Rev: V0.1**
KiCad E.D.A. 10.0.1		Id: 1/6



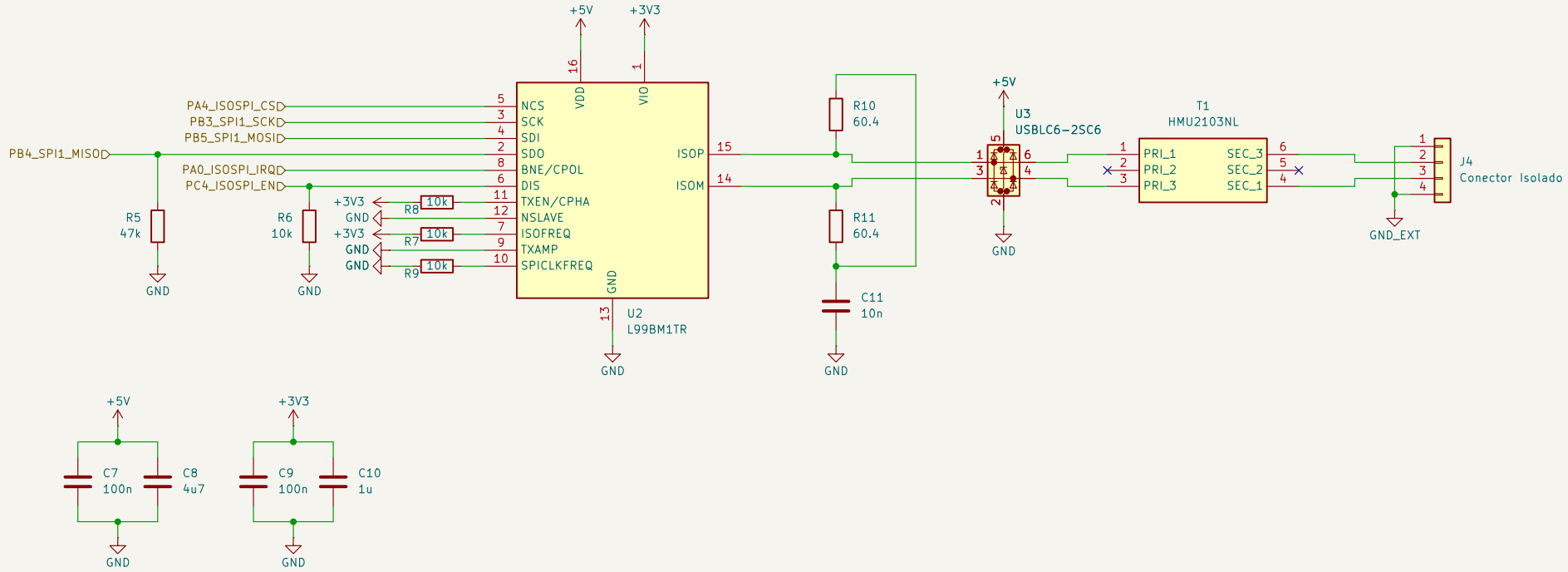
Buck Síncrono Automotivo 5.0V / 1.5A contínuos
TVS Automotivo 1500W SMBJ15CA e Filtro LC em Pi
Proteção contra reversão de polaridade (Schottky)

Formula ITA

Sheet: /01_Power_Supply_GLV/
File: 01_power_supply.kicad_sch

Title: Bloco 01 – Regulador Automotivo GLV 12V -> 5V (ISO 7637-2)

Size: A4	Date: 2026-10-22	**Rev: V0.1**
KiCad E.D.A. 10.0.1		Id: 2/6



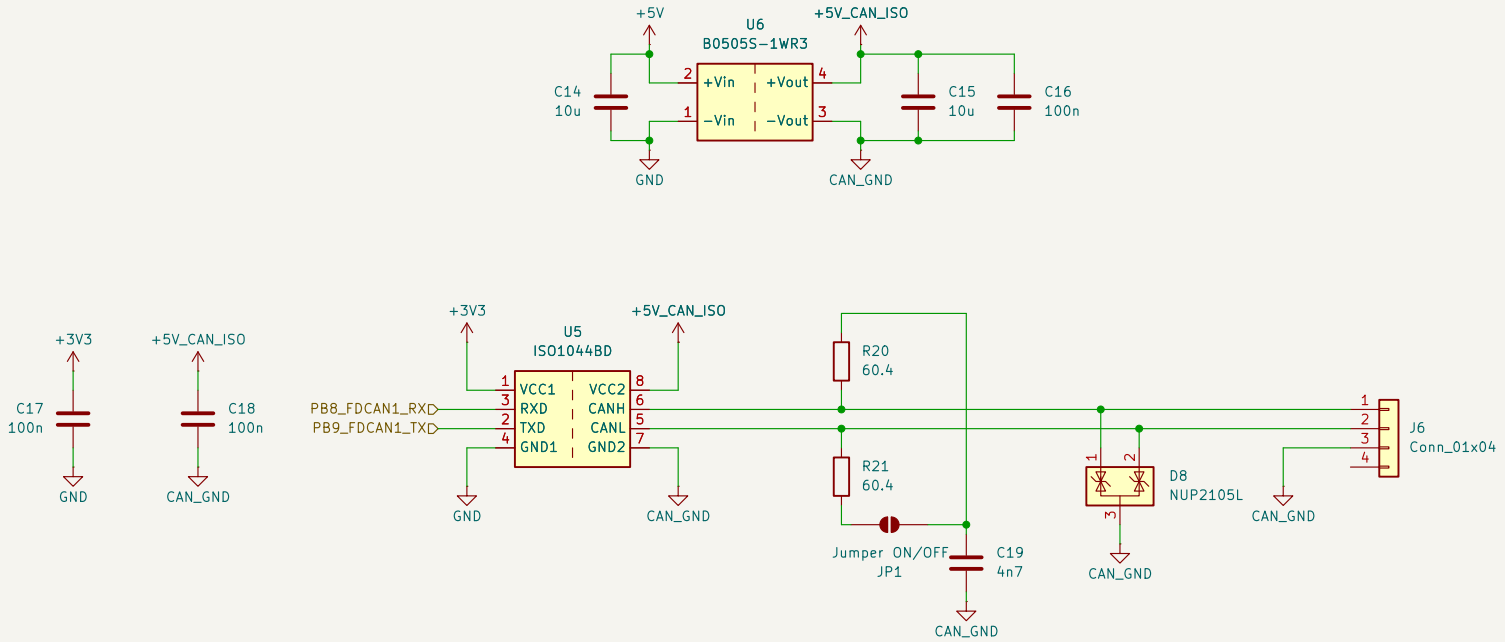
Zona de exclusão de cobre (Keepout) com creepage >= 4.0 mm
Conector Molex Micro-Fit 3.0 de 4 pinos para os Slaves 5S
Transformador de pulso Pulse HMU2103NL (Isolamento >= 2500V RMS)
Transceptor Iso-SPI automotivo ST L99BM1TR (SO-16N)

Formula ITA

Sheet: /02_IsoSPI_Galvanic_Barrier/
File: 02_isospi_barrier.kicad_sch

Title: Bloco 02 – Subsistema Iso-SPI com Barreira Galvânica 2500V

Size: A4	Date: 2026-10-22	**Rev: V0.1**
KiCad E.D.A. 10.0.1		Id: 4/6



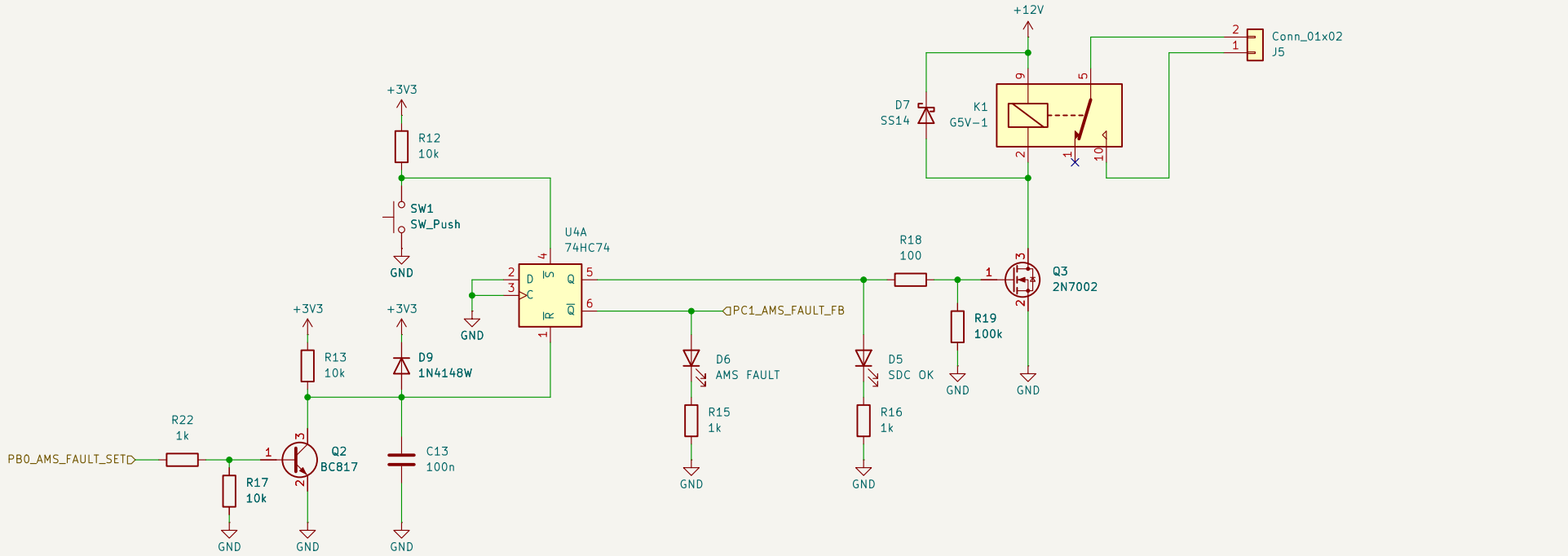
Resistor de terminação selecionável por jumper de 120 Ohms com choke
Taxa de telemetria até 5 Mbps com barreira capacitiva de alta imunidade
Transceptor FDCAN Isolado TI ISO1042 (Isolamento 5000V RMS)

Formula ITA

Sheet: /04_Isolated_CANFD_Interface/
File: 05_canfd_interface.kicad_sch

Title: Bloco 05 – Transceptor FDCAN Isolado e Barramento Veicular

Size: A4	Date: 2026-10-22	**Rev: V0.1**
KiCad E.D.A. 10.0.1		Id: 5/6



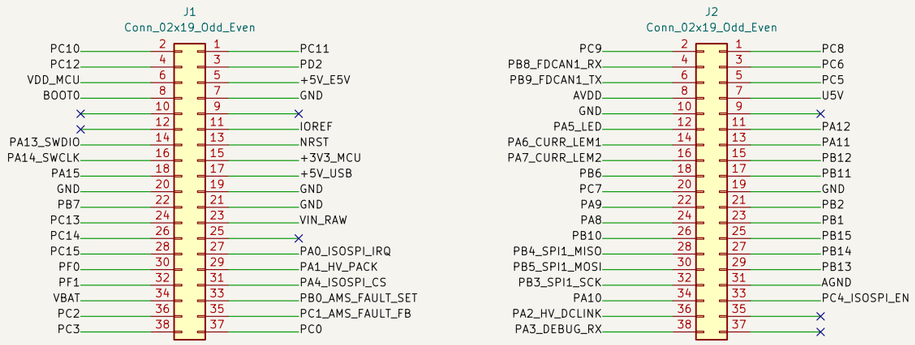
Botão de Reset manual de baixa tensão para rearme em box
Disparo instantâneo sob pulso de microsegundos do MCU
Relé de Segurança normalmente aberto (NO) no loop SDC
Trava biestável por hardware (Flip-Flop / SCR equivalente)

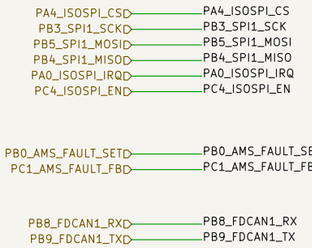
Formula ITA

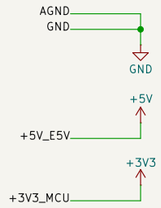
Sheet: /03_SDC_Safety_Latch/
File: 03_sdc_latch.kicad_sch

Title: Bloco 03 – Circuito Latch de Falha AMS & Relé SDC (FSAE EV4.

Size: A4	Date: 2026-10-22	**Rev: V0.1**
KiCad E.D.A. 10.0.1		Id: 6/6









Pinout verificado conforme User Manual ST UM2505 / UM1724
Alimentação via pino E5V (5V regulado da Carrier Board)
Conexão oficial padronizada ST Morpho (2x 2x19 pinos)
Formula ITA — Formula SAE Electric

Sheet: /05_Nucleo_G474RE_Morpho_Headers/
File: 06_nucleo_headers.kicad_sch

Title: Bloco 06 — Headers Morpho ST CN7 e CN10 (NUCLEO-G474RE)

Size: A3	Date: 2026-10-22	**Rev: V0.1**
KiCad E.D.A. 10.0.1		Id: 7/6